

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I _____meghanadh a (192372281)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *meghanadh*

Annexure A Scenario-based

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

3A)

Key Exchange in an Unsecured Network

A. How the Diffie–Hellman Key Exchange Works

Diffie–Hellman (DH) allows two users to create a shared secret key over a public network without sending the secret key directly.

Steps:

1. Both users publicly agree on:
 - A large prime number (**P**)
 - A generator (**G**)
2. **User A** chooses a private key **a** and calculates:
 - **$A = G^a \bmod P$**
 - Sends **A** to User B.
3. **User B** chooses a private key **b** and calculates:
 - **$B = G^b \bmod P$**
 - Sends **B** to User A.
4. Both users compute the shared secret:
 - User A: **$K = B^a \bmod P$**
 - User B: **$K = A^b \bmod P$**
5. Since both calculations produce the same value, both users now share the **same secret key**, which can be used for secure encryption and decryption.

B. Security Threat and Solution

Possible Security Threat:

Man-in-the-Middle (MITM) Attack

- An attacker intercepts the public keys exchanged between the two users.
- The attacker establishes separate keys with each user.
- Both users believe they are communicating securely with each other, but the

attacker can read and modify all messages.

Solution to Mitigate the Threat:

- Use **authentication** along with Diffie–Hellman.
- Authenticate the exchanged public keys using:
 - **Digital Certificates**
 - **Digital Signatures**
 - Secure protocols such as **TLS (Transport Layer Security)**

This ensures that the public keys belong to the intended users and prevents attackers from impersonating either party.

Conclusion

Diffie–Hellman enables two users to establish a shared secret key over an unsecured network. However, it is vulnerable to Man-in-the-Middle attacks if authentication is not used. Combining Diffie–Hellman with digital certificates or TLS provides secure and authenticated key exchange.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

4A)

Public Key Cryptography for Email Security.

A. How RSA is Used for Encryption and Decryption

RSA is a **public key cryptography** algorithm used to provide secure email communication.

Steps:

1. Each user generates two keys:
 - **Public Key** – Shared with others.
 - **Private Key** – Kept secret.
2. When **User A** wants to send a secure email to **User B**:
 - User A encrypts the email using **User B's public key**.

- The encrypted email is sent over the network.
- 3. When **User B** receives the email:
 - User B decrypts it using **their private key**.
 - Since only User B has the private key, only they can read the message.

Result:

The email remains confidential even if it is intercepted during transmission.

B. Role of Key Distribution and Management

Key distribution and management are essential for maintaining the security of RSA.

- **Public Key Distribution:** Public keys must be shared securely so users can encrypt emails for the correct recipient.
- **Private Key Protection:** Private keys must remain confidential and should never be shared.
- **Digital Certificates:** Certificates issued by a **Certificate Authority (CA)** verify that a public key belongs to the correct user, preventing impersonation attacks.
- **Key Renewal and Revocation:** Keys should be updated periodically, and compromised or expired keys must be revoked and replaced.
- **Secure Key Storage:** Private keys should be stored securely, preferably using encryption or hardware security modules (HSMs).

Conclusion

RSA secures email communication by encrypting messages with the recipient's **public key** and decrypting them with the corresponding **private key**. Proper key distribution, secure storage, digital certificates, and regular key management ensure the confidentiality and authenticity of email communications.

RUBICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand